EOSC 213: Computational Methods in Geological Engineering

Lecture 13: Boundary Value Problems (BVPs) + Finite Differences for BVPs

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Correction: Eigenvalues of triangular matrices

Both **upper** and **lower** triangular matrices have eigenvalues equal to their diagonal entries.

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Observe, the transpose (rows $\leftrightarrow$ columns) of L is upper-triangular,

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Key facts

- $\det(A) = \det(A^\top)$

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Therefore,

$$\det(L - \lambda I) = \det(L^\top - \lambda I),$$

so the eigenvalues of L are also its diagonal entries.

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Triangular matrices reveal their eigenvalues directly on the diagonal.

Logistics: Midterm I solutions

Midterm I solutions are now posted on Canvas.

Please:

- Carefully review the full solutions.
- Compare with your own approach — not just final answers.
- Identify where mistakes occurred:
  - conceptual understanding,
  - algebra/calculation errors,
  - numerical method setup.

Goal: use the midterm as feedback for improvement before Midterm II and the Final Exam.

Logistics: TA Office Hours + Course Feedback Survey

TA Office Hours Update

To better match student demand:

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Mid-Course Student Feedback

- Please fill out the short survey posted on Canvas.
- Your responses will help us improve the course content as well as how it is delivered and your experience taking this course.

Recap: What is an ODE?

An **ordinary differential equation (ODE)** relates a function and its derivatives,

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Goal: find $y(t)$ either **analytically** (if possible, usually it is not) or **numerically** (what we learn in this course).

Recap: Initial Value Problems (IVPs)

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Key idea:

- The solution is fully determined by the initial state.
- We compute the solution by moving forward in time.

Known start $\Rightarrow$ evolve forward.

Recap: Numerical time-stepping methods

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All are based on approximating derivatives locally.

Recap: Accuracy and stability of numerical methods

We learned that numerical methods must satisfy:

Accuracy

- local truncation error
- order of convergence

Stability

- small errors should not grow uncontrollably
- depends on step size h

Recap: Systems of ODEs

Many physical problems involve multiple variables,

$$\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}).$$

We rewrote systems using matrices,

$$\mathbf{y}' = A\mathbf{y}.$$

Key concepts:

- coupled dynamics
- eigenvalues and eigenvectors
- phase-space intuition

Recap: Implementation viewpoint

Numerical solution workflow:

1. Choose time grid
2. Approximate derivatives
3. Update solution step-by-step
4. Analyze stability and behavior

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Important observation:

All numerical methods that we discussed so far compute the solution *sequentially*.

$$y_0 \to y_1 \to y_2 \to \cdots$$

Recap: ODEs as *time evolution* (IVPs)

What we have done so far,

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Key feature: the solution is built *sequentially*.

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You are told information at multiple locations:

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These are called **Boundary Value Problems (BVPs)**.

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Question: How do we "march forward" if we are constrained at *both ends*?

Answer: We usually cannot. We must solve **globally**.

IVP vs BVP (big picture)

Initial Value Problem (IVP)	**Boundary Value Problem (BVP)**
Given at one point (usually "start time")	Given at multiple points (often boundaries)
Sequential updates (marching)	Global coupling (solve all at once)
Natural for dynamics in time	Natural for equilibrium / steady-state / spatial profiles

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Important: We will study BVPs mostly for **second-order ODEs** since they naturally arise in physics/geophysics. But, BVPs are *not* only for second-order ODEs. They can appear for first-order ODEs too (less common in physics, but valid).

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Common boundary types:

- **Dirichlet:** $y(0) = a$, $y(L) = b$ (value prescribed at the boundary)
- **Neumann:** $y'(0) = \alpha$, $y'(L) = \beta$ (derivative/slope prescribed at the boundary)
- **Mixed/Robin:** $\gamma y + \delta y' = r$ at the boundary

Physical motivations (why BVPs show up)

BVPs arise naturally when the solution must satisfy **constraints at boundaries**.

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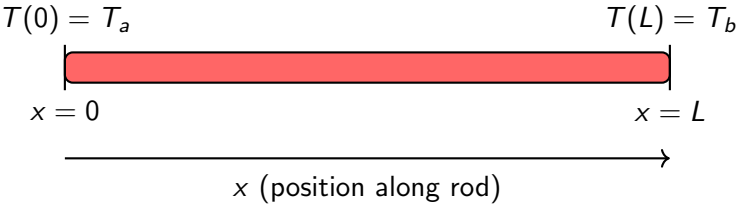
Key point: these are often *equilibrium / steady* problems.

Example 1: Steady heat in 1D (with Dirichlet BC)



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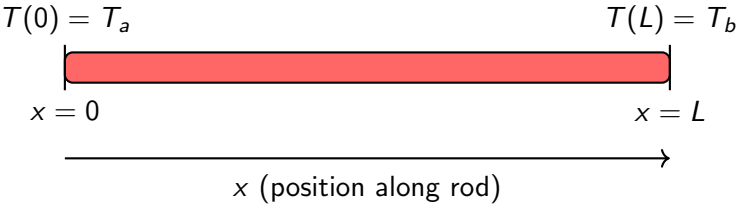

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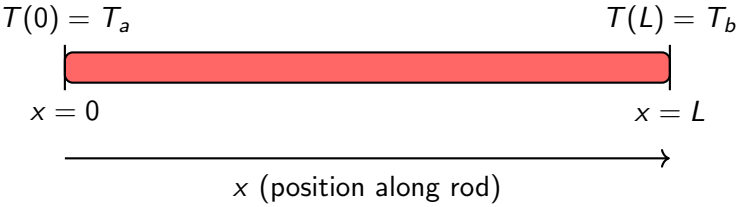

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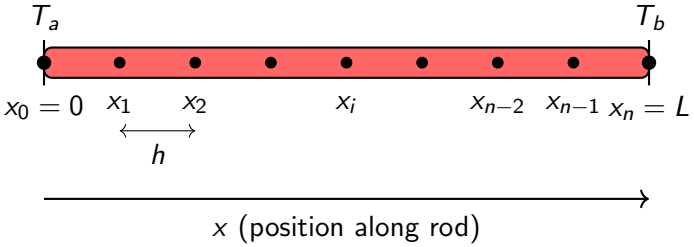
Boundary conditions (temperature fixed): $T(0) = T_a, \quad T(L) = T_b$.
This is a **BVP** because constraints are at both ends.

Discretize the domain



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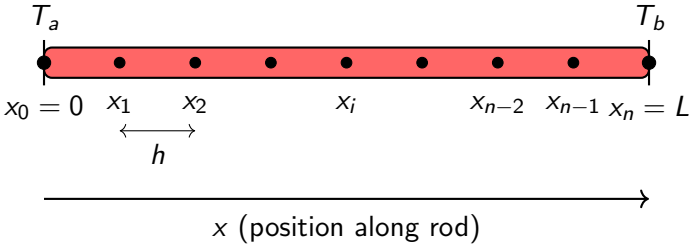

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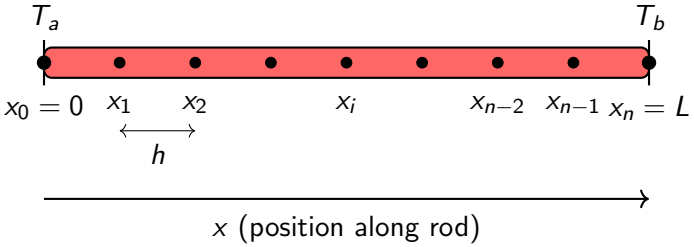

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Boundary values are known: $T_0 = T_a, \quad T_n = T_b$.
Unknowns are the **interior points:** $T_1, \ldots, T_{n-1}$.

Finite differences: first vs second derivatives

First derivative (used earlier for IVPs)

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For interior points $i = 1, \ldots, n-1$:

$$T''(x_i) \approx \frac{T_{i+1} - 2T_i + T_{i-1}}{h^2}.$$

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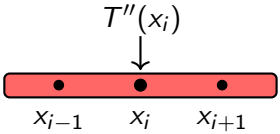
Note: *Second derivative uses 3 points (3-point stencil).*



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Plug into $-kT''(x) = q(x)$,

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Rearrange,

$$-T_{i-1} + 2T_i - T_{i+1} = \frac{h^2}{k}\, q(x_i).$$

Key shift: the unknowns are *coupled*

Notice what happened:

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So we do NOT compute T_1 then T_2 then T_3 by marching (like we used to do for IVPs).

Instead, we will solve a **linear system** for all interior values at once.

From finite differences to a linear system (start with equations)

For each interior grid point $i = 1, \ldots, n-1$, we have,

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Pattern: each equation couples a point with its two neighbors.

Separate known vs unknown: interior temperatures

Boundary temperatures are known,

$$T(x_0 = 0) = T_0 = T_a, \qquad T(x_n = L) = T_n = T_b.$$

Separate known vs unknown: interior temperatures

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$$T(x_0 = 0) = T_0 = T_a, \qquad T(x_n = L) = T_n = T_b.$$

Unknowns are the **interior** values,

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Stack the interior unknowns into a vector,

$$\mathbf{T} = \begin{bmatrix} T_1 \\ T_2 \\ \vdots \\ T_{n-1} \end{bmatrix}.$$

We will solve for all interior temperatures at once using matrix form.

Boundary conditions move to the right-hand side

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Message: boundary values become part of the forcing term.

Stack all equations: $A\mathbf{T} = \mathbf{b}$

Writing all interior equations together gives a linear system,

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Matrix A

$$\begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}$$

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$$\begin{bmatrix} T_1 \\ T_2 \\ \vdots \\ T_{n-1} \end{bmatrix}$$

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Each row of A corresponds to one interior grid point equation.

Solving $A\mathbf{T} = \mathbf{b}$ using PyTorch

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- A encodes how grid points interact
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Physics → discretization → linear algebra → solution

Example 2: Steady heat in 1D (with Neumann BC)

Same ODE as Example 1,

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Physical interpretation

Neumann BCs specify **heat flux** at the boundaries (insulated / imposed flux), instead of fixing the temperature values.

Discretize the rod (same grid as before)

Take the same grid points:

$$x_i = ih, \qquad i = 0, 1, \ldots, n, \qquad h = \frac{L}{n}.$$

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- Now we do *not* know T_0 or T_n; instead we know the slopes.

So we keep **all** nodal values as unknowns:

$$T_0,\ T_1,\ \ldots,\ T_n.$$

Interior finite difference equations (unchanged)

For interior points $i = 1, \ldots, n-1$ (same as Example 1):

$$T''(x_i) \approx \frac{T_{i+1} - 2T_i + T_{i-1}}{h^2}.$$

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Message: only the *boundary treatment* changes.

Finite differences for Neumann boundary conditions

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Left boundary ($x = 0$):

$$T'(0) = \alpha \quad \approx \quad \frac{T_1 - T_0}{h} = \alpha \quad \Rightarrow \quad -T_0 + T_1 = h\alpha.$$

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Boundary conditions become equations

Instead of plugging known values into **b**, we add two new linear equations.

Linear system structure (what changes vs Dirichlet?)

Unknown vector now includes all grid values,

$$\mathbf{T} = \begin{bmatrix} T_0 \\ T_1 \\ \vdots \\ T_{n-1} \\ T_n \end{bmatrix}.$$

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But the **first and last rows** come from Neumann BCs:

$$-T_0 + T_1 = h\alpha, \qquad -T_{n-1} + T_n = h\beta.$$

Same building blocks as Example 1, just different boundary rows.

Linear system structure (what changes vs Dirichlet?)

$$A\mathbf{T} = \mathbf{b}.$$

All nodal temperatures are now unknown,

Matrix A

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Important note

Pure Neumann problems

If both boundaries are Neumann, the solution is determined only up to an additive constant (adding a constant does not change derivatives).

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If both boundaries are Neumann, the solution is determined only up to an additive constant (adding a constant does not change derivatives).

In practice, we fix a reference (e.g., specify $T(0)$ or the mean temperature) to make the solution unique.

Mini takeaways

- BVP $\Rightarrow$ we discretized space (not time)
- Finite differences convert derivatives into linear algebra
- BVP discretization typically gives a **system** of equations
- This is why linear algebra matters in this course

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We have turned calculus into a matrix equation.

Common misconception: BVPs are **not** only for second-order ODEs

Example of a first-order BVP,

$$y'(x) = f(x), \qquad y(0) = a, \quad y(L) = b.$$

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Why second order is common in physics: many equilibrium laws involve curvature (second derivative) and fluxes.

Summary: what to remember

Big idea: Boundary Value Problems (BVPs)

Today we learned how spatial differential equations are solved computationally.

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1. Physical model $\rightarrow$ differential equation
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4. Obtain a linear system $A\mathbf{T} = \mathbf{b}$
5. Solve using numerical linear algebra tools

$$\textit{Calculus problem} \Rightarrow \textit{Linear algebra problem}$$

Next lecture: From BVPs to Implicit Methods

So far:

- IVPs were solved by stepping forward in time
- BVPs required solving a globally coupled system of equations

Next question:

What happens when time-stepping methods also become globally coupled?

Next lecture(s):

- Motivation for implicit methods
- Stability limitations of explicit methods
- Stiff differential equations
- Backward Euler method
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Implicit methods connect time-stepping with ideas we saw in BVPs.